MATH 180 Strategies of Problem Solving

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Outline

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Chomp Review

Last time

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- we learned the "strategy-stealing argument"

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- we proved Player 1 has a perfect strategy

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- we learned the "strategy-stealing argument"
- we proved Player 1 has a perfect strategy
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- by **symmetry**: strategy for $n \times 1$ and $n \times 2$ Chomp

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- we learned the "strategy-stealing argument"
- we proved Player 1 has a perfect strategy
- we figured out the strategy for $1 \times n$ and $2 \times n$ Chomp
- by **symmetry**: strategy for $n \times 1$ and $n \times 2$ Chomp

Question

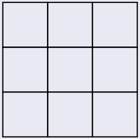
What about $3 \times n$ chomp?

Perfect Strategy

Perfect Strategy

Question

What is the perfect strategy for 3×3 Chomp?

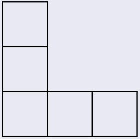


Perfect Strategy

Perfect Strategy

Answer

First move:



Perfect Strategy

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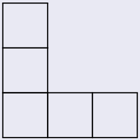
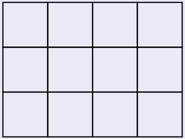


Then Copycat Strategy!

Perfect Strategy

Question

What is the best first move for 3×4 Chomp?



Perfect Strategy

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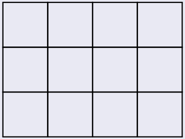


Activity (10 mins)

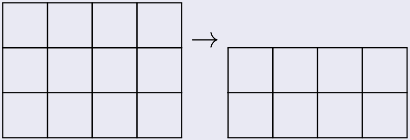
Break into small groups and discuss. What's the strategy?

Perfect Strategy

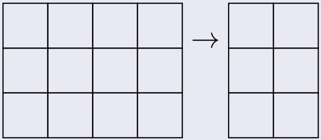
First move?

Can't be left column or bottom row, because it'd make a rectangle (winning board), eg.

$(1, 1)$:



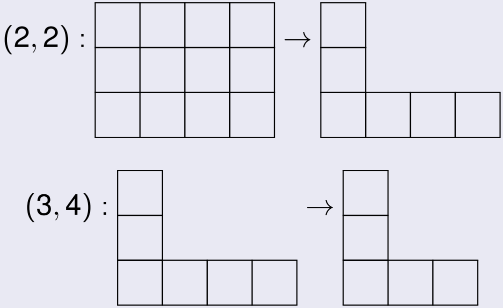
$(3, 3)$:



Perfect Strategy

First move?

Can't be $(2,2)$, since then they could follow up with $(3,4)$



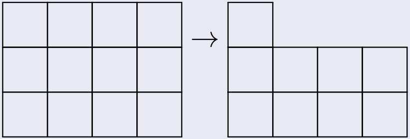
Then we lose to Copycat Strategy

Perfect Strategy

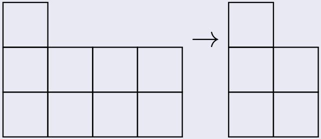
First move?

Can't be $(1, 2)$, since then they could follow up with $(3, 3)$

$(1, 2)$:



$(3, 3)$:



Then we lose to the $n \times 2$ strategy

Perfect Strategy

Theorem

The best first move for 3×4 Chomp is $(2, 3)$.

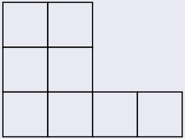
Perfect Strategy

Theorem

The best first move for 3×4 Chomp is $(2, 3)$.

Proof

Board after the first move:



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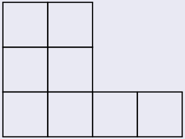
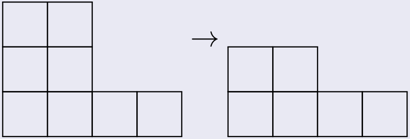


How could Player 2 follow up?

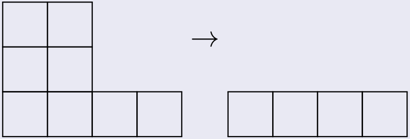
Perfect Strategy

Cases

$(1, 1)$:



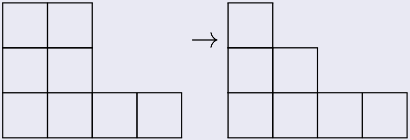
$(2, 1)$:



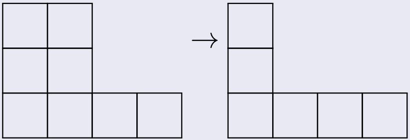
Perfect Strategy

Cases

$(1, 2)$:



$(2, 2)$:



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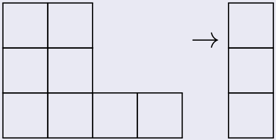
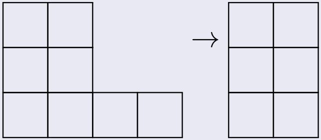


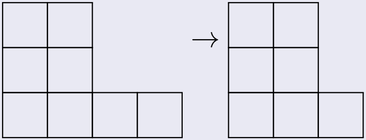
Perfect Strategy

Cases

$(3,3)$:



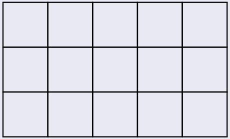
$(3,4)$:



Perfect Strategy

Question

What is the best first move for 3×5 Chomp?



Activity (20 mins)

Break into small groups and discuss. How many moves can you eliminate?

Perfect Strategy

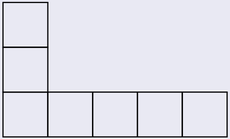
Process of elimination

X				
X				
X	X	X	X	X

Avoid left side and bottom due to rectangles.

Perfect Strategy

What about $(2, 2)$?



Perfect Strategy

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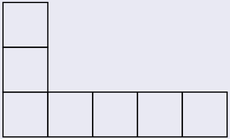


Then they could just do the Copycat Strategy.

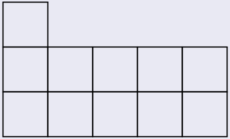
Perfect Strategy

Process of elimination

X				
X	X			
X	X	X	X	X

Perfect Strategy

What about $(1, 2)$?



Perfect Strategy

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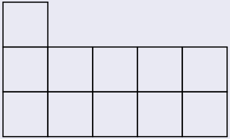
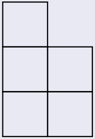


They can follow up with a $(3, 3)$



and then use $n \times 2$ tactics

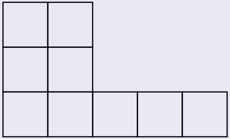
Perfect Strategy

Process of elimination

X	X			
X	X			
X	X	X	X	X

Perfect Strategy

What about $(2, 3)$?



Perfect Strategy

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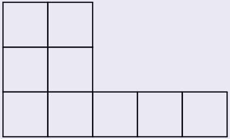
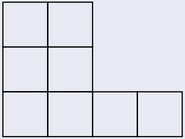


They can follow up with a $(3, 5)$



Looks like they did the winning move for 3×4

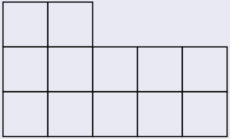
Perfect Strategy

Process of elimination

X	X			
X	X	X		
X	X	X	X	X

Perfect Strategy

What about $(1, 3)$?



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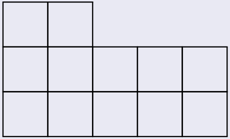
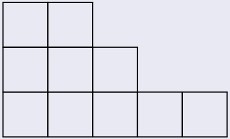


They can follow up with a $(2, 4)$

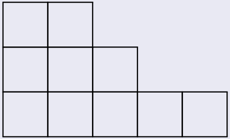


This is a losing board!

Prove it

Activity (10 mins)

Prove that the board



is a **losing board**, ie. the next player to play can be forced to lose.

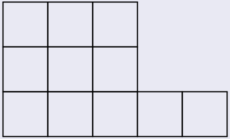
Perfect Strategy

Process of elimination

X	X	X		
X	X	X		
X	X	X	X	X

Perfect Strategy

What about $(2, 4)$?



Perfect Strategy

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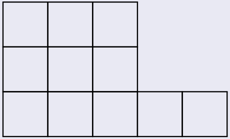

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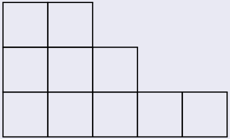


We just saw this was a losing position.

Perfect Strategy

Process of elimination

X	X	X		
X	X	X	X	
X	X	X	X	X

Perfect Strategy

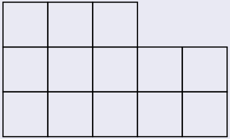
Process of elimination

x	x	x	?	?
x	x	x	x	?
x	x	x	x	x

Perfect Strategy

Theorem

The perfect first move for 3×5 Chomp is $(1, 4)$.



Outline

Getting moves for higher n values

Finding the perfect move requires ...

Getting moves for higher n values

Finding the perfect move requires ...

- remembering lots of past boards

Getting moves for higher n values

Finding the perfect move requires ...

- remembering lots of past boards
- checking lots of “if then" cases

Getting moves for higher n values

Finding the perfect move requires ...

- remembering lots of past boards
- checking lots of “if then" cases

Computers are really good at this!

Computer calculating of best $3 \times n$ move

n	Width w	Play at (row, col)
2	1	(1,2)
3	2	(2,2)
4	2	(2,3)
5	2	(1,4)
6	3	(2,4)
7	3	(1,5)
8	4	(2,5)
9	3	(1,7)
10	5	(2,6)
11	5	(2,7)
12	4	(1,9)
13	6	(2,8)
14	4	(1,11)
15	7	(2,9)

Positions of Best First Moves

	2		5	7		9		12		14
1	3	4	6	8	10	11	13	15	16	18

Entries represent the board length for which the corresponding spot is the best first move.

Question

Can you see any patterns?

Notice patterns

Submit your findings:

Create conjectures

Conjecture 1

Each intege n end up in a unique spot in the table.

Create conjectures

Conjecture 1

Each intege *n* end up in a unique spot in the table.

Conjecture 2

The numbers in any row get bigger from left to right

Create conjectures

Conjecture 1

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Conjecture 2

The numbers in any row get bigger from left to right

Conjecture 3

The second row doesn’t have any “gaps"

Create conjectures

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The second row doesn’t have any “gaps"

Next steps

Create conjectures

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Conjecture 2

The numbers in any row get bigger from left to right

Conjecture 3

The second row doesn't have any “gaps"

Next steps

- gather more evidence

Create conjectures

Conjecture 1

Each intege *n* end up in a unique spot in the table.

Conjecture 2

The numbers in any row get bigger from left to right

Conjecture 3

The second row doesn’t have any “gaps"

Next steps

- gather more evidence
- try to prove it!

Computer calculating of best $3 \times n$ move

n	Play at (row, col)
2	(1 , 2)
3	(2 , 2)
4	(2 , 3)
5	(1 , 4)
6	(2 , 4)
7	(1 , 5)
8	(2 , 5)
9	(1 , 7)
10	(2 , 6)
11	(2 , 7)
12	(1 , 9)
13	(2 , 8)
14	(1 , 11)

Computer calculating of best $3 \times n$ move

n	Play at (row, col)
15	(2 , 9)
16	(2 , 10)
17	(1 , 13)
18	(2 , 11)
19	(1 , 14)
20	(2 , 12)
21	(2 , 13)
22	(1 , 16)
23	(1 , 17)
24	(2 , 14)
25	(2 , 15)
26	(1 , 19)
27	(2 , 16)

Computer calculating of best 3 × n move

n	Play at (row, col)
28	(2 , 17)
29	(1 , 21)
30	(2 , 18)
31	(1 , 22)
32	(2 , 19)
33	(1 , 24)
34	(2 , 20)
35	(2 , 21)
36	(1 , 26)
37	(2 , 22)
38	(1 , 28)
39	(2 , 23)
40	(2 , 24)

Computer calculating of best $3 \times n$ move

n	Play at (row, col)
41	(1 , 30)
42	(2 , 25)
43	(1 , 31)
44	(2 , 26)
45	(2 , 27)
46	(1 , 33)
47	(1 , 34)
48	(2 , 28)
49	(2 , 29)
50	(1 , 36)
51	(2 , 30)
52	(1 , 38)

Computer calculating of best $3 \times n$ move

n	Play at (row, col)
53	(2 , 31)
54	(2 , 32)
55	(1 , 40)
56	(2 , 33)
57	(2 , 34)
58	(1 , 42)
59	(2 , 35)
60	(2 , 36)
61	(1 , 44)
62	(1 , 45)
63	(2 , 37)
64	(2 , 38)
65	(1 , 47)

Prove conjectures

Theorem

Each intege n end up in a unique spot in the table.

Prove conjectures

Theorem

Each intege n end up in a unique spot in the table.

Proof

Assume there is a spot (j, k) which occurs twice, for values m, n with $0 < m < n$.

Prove conjectures

Theorem

Each intege n end up in a unique spot in the table.

Proof

Assume there is a spot (j, k) which occurs twice, for values m, n with $0 < m < n$.
Start with a $3 \times n$ grid.

Prove conjectures

Theorem

Each intege n end up in a unique spot in the table.

Proof

Assume there is a spot (j, k) which occurs twice, for values m, n with $0 < m < n$.
Start with a $3 \times n$ grid.
Then if Player 1 chooses (j, k) they should win.

Prove conjectures

Theorem

Each intege n end up in a unique spot in the table.

Proof

Assume there is a spot (j, k) which occurs twice, for values m, n with $0 < m < n$.
Start with a $3 \times n$ grid.
Then if Player 1 chooses (j, k) they should win.
However, if Player 2 follows by choosing $(3, n - m + 1)$, it's equivalent to them having choosen (j, k) on the $3 \times n$ grid.

Prove conjectures

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Therefore Player 2 should be the one that wins.

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Start with a $3 \times n$ grid.
Then if Player 1 chooses (j, k) they should win.
However, if Player 2 follows by choosing $(3, n - m + 1)$, it's equivalent to them having choosen (j, k) on the $3 \times n$ grid.
Therefore Player 2 should be the one that wins.
Contradiction.

Or find a counter-example

n	Play at (row, col)
66	(2 , 39)
67	(1 , 48)
68	(2 , 40)
69	(1 , 50)
70	(2 , 41)
71	(1 , 51)
72	(2 , 42)
73	(2 , 43)
74	(2 , 44)
75	(1 , 53)
76	(2 , 45)
77	(1 , 55)
78	(2 , 46)

Computer calculating of best $3 \times n$ move

n	Play at (row, col)
79	(1 , 56)
80	(2 , 47)
81	(1 , 58)
82	(2 , 48)
83	(2 , 49)
84	(1 , 60)
85	(2 , 50)
86	(1 , 62)
87	(2 , 51)
88	(2 , 53)
89	(2 , 52)
90	(1 , 64)
91	(1 , 65)

Computer calculating of best $3 \times n$ move

n	Play at (row, col)
92	(2 , 54)
93	(2 , 55)
94	(1 , 67)
95	(2 , 56)
96	(2 , 57)
97	(1 , 69)
98	(2 , 58)
99	(1 , 71)

Counter-example

Conjecture

The number in the middle row must increase.

Counter-example

Conjecture

The number in the middle row must increase.

First counter-example

For $n = 88$, the best first move is $(2, 53)$, but for $n = 87$ the best first move is $(2, 52)$

Strong Law of Small Numbers

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Richard Guy:

There aren't enough small numbers to make the many demands of them.

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Meaning

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- Mathematics is full of conjectures which are true for small numbers, but then fail for larger ones.

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- Mathematics is full of conjectures which are true for small numbers, but then fail for larger ones.
- Observations are evidence, but not proofs!

Strong Law of Small Numbers

Richard Guy:

There aren't enough small numbers to make the many demands of them.

Meaning

- Mathematics is full of conjectures which are true for small numbers, but then fail for larger ones.
- Observations are evidence, but not proofs!
- Be suspicious!

Strong Law of Small Numbers

Example (Primes)

Strong Law of Small Numbers

Example (Primes)

- 31 is prime

Strong Law of Small Numbers

Example (Primes)

- 31 is prime
- 331 is prime

Strong Law of Small Numbers

Example (Primes)

- 31 is prime
- 331 is prime
- 3331 is prime

Strong Law of Small Numbers

Example (Primes)

- 31 is prime
- 331 is prime
- 3331 is prime
- 33331 is prime

Strong Law of Small Numbers

Example (Primes)

- 31 is prime
- 331 is prime
- 3331 is prime
- 33331 is prime
- 333331 is prime

Strong Law of Small Numbers

Example (Primes)

- 31 is prime
- 331 is prime
- 3331 is prime
- 33331 is prime
- 333331 is prime
- 3333331 is prime

Strong Law of Small Numbers

Example (Primes)

- 31 is prime
- 331 is prime
- 3331 is prime
- 33331 is prime
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- 33333331 is prime

Strong Law of Small Numbers

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Strong Law of Small Numbers

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$$333333331 = 17 \times 19607843$$

Strong Law of Small Numbers

Example (Diophantine Equations)

For a long time, it was conjectured that

$$x^4 + y^4 + z^4 = w^4$$

has no nonzero **integer-valued** solutions.

Strong Law of Small Numbers

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First counter-example

$$95800^4 + 217519^4 + 414560^4 = 422481^4$$

Strong Law of Small Numbers

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First counter-example

$$n = 906150257$$

Strong Law of Small Numbers

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Example (Goldbach Conjecture)

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$12 = 7 + 5$, $30 = 13 + 17$, $46 = 23 + 23$.

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Checked up through:

$$n = 4,000,000,000,000,000,000$$

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Checked up through:

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no known counter-examples

Strong Law of Small Numbers

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Example (Collatz Conjecture)

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$f(x) = 3x + 1$ if x is odd

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Strong Law of Small Numbers

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$$n = 295, 147, 905, 179, 352, 825, 856$$

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Outline

1. SoPS Lecture 4
   - $3 \times n$ Chomp
   - Strong Law of Small Numbers
   - Wrapping up

Discussion

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Conjecture vs. Proof

In science evidence is enough. In mathematics, we require proof because evidence can mislead us.

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Question 1

Is the best first move on a $3 \times n$ easy to predict?

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Conjecture vs. Proof

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Question 1

Is the best first move on a $3 \times n$ easy to predict?

Question 2

What does the Strong Law of Small Numbers mean in your own words? Can you think of any examples of the Strong Law of Small Numbers in action?

Question of the Day

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Puzzle

Suppose you start with a 3×4 board. How many possible board shapes can we make by doing only Chomp moves.

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Activity (20 mins)

Discuss the puzzle with your neighbors. What is the solution?

Final thoughts

Reflection

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- Looking at the Chomp data above, come up with your own Conjecure about the best first move on a $3 \times n$ Chomp board.

Final thoughts

Reflection

- Looking at the Chomp data above, come up with your own Conjecure about the best first move on a $3 \times n$ Chomp board.
- Describe the Strong Law of Small Numbers in your own terms. What does it say about evidence in mathematics. Does it mean that evidence is useless?

Submit your responses by **Tonight!**